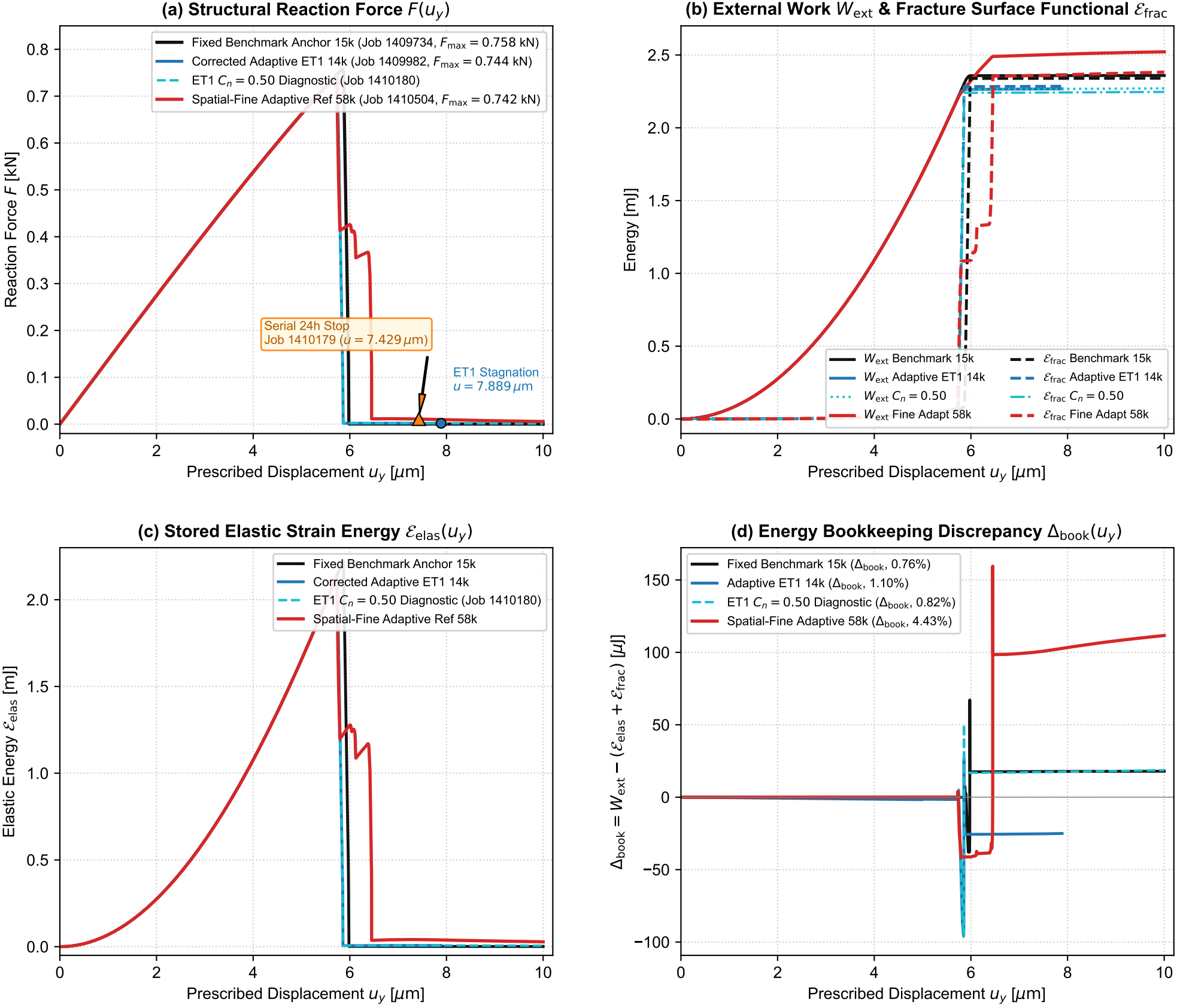

(a) Structural Reaction Force $F(u_y)$
Fixed Benchmark Anchor 15k (Job 1409734, F_{max} = 0.758 kN)
Corrected Adaptive ET1 14k (Job 1409982, F_{max} = 0.744 kN)
ET1 C_n = 0.50 Diagnostic (Job 1410180)
Spatial-Fine Adaptive Ref 58k (Job 1410504, F_{max} = 0.742 kN)
Serial 24h Stop
Job 1410179 (u = 7.429 μm)
ET1 Stagnation
u = 7.889 μm
Reaction Force F [kN]
Prescribed Displacement u_y [μm]
(b) External Work W_{ext} & Fracture Surface Functional $\mathcal{E}_{frac}$
W_{ext} Benchmark 15k
W_{ext} Adaptive ET1 14k
W_{ext} C_n = 0.50
W_{ext} Fine Adapt 58k
$\mathcal{E}_{frac}$ Benchmark 15k
$\mathcal{E}_{frac}$ Adaptive ET1 14k
$\mathcal{E}_{frac}$ C_n = 0.50
$\mathcal{E}_{frac}$ Fine Adapt 58k
Energy [mJ]
Prescribed Displacement u_y [μm]
(c) Stored Elastic Strain Energy $\mathcal{E}_{elas}(u_y)$
Fixed Benchmark Anchor 15k
Corrected Adaptive ET1 14k
ET1 C_n = 0.50 Diagnostic (Job 1410180)
Spatial-Fine Adaptive Ref 58k
Elastic Energy $\mathcal{E}_{elas}$ [mJ]
Prescribed Displacement u_y [μm]
(d) Energy Bookkeeping Discrepancy $\Delta_{book}(u_y)$
Fixed Benchmark 15k (Δ_{book}, 0.76%)
Adaptive ET1 14k (Δ_{book}, 1.10%)
ET1 C_n = 0.50 Diagnostic (Δ_{book}, 0.82%)
Spatial-Fine Adaptive 58k (Δ_{book}, 4.43%)
$\Delta_{book} = W_{ext} - (\mathcal{E}_{elas} + \mathcal{E}_{frac})$ [μJ]
Prescribed Displacement u_y [μm]